



Fundamentals of Database System

Chapter 4: Functional Dependency and Normalization

Compiled by: Diriba G (MSc. in CSE)

Outline

- ❖ Overview
- ❖ Functional dependency
- ❖ Normal forms

Functional Dependency

- ❖ **Data Dependency:** Two data items A and B are said to be in a determinant or dependent relationship if certain values of data item B always appears with certain values of data item A.
- ❖ The essence of this idea is that **if the existence of something, call it A, implies that B must exist and have a certain value, then we say that “B is functionally dependent on A.”**
- ❖ **Functional dependency(*FD*):** is a set of constraints between two attributes in a relation. Functional dependency says that if two tuples have same values for attributes A1, A2,..., An, then those two tuples must have to have same values for attributes B1, B2, ..., Bn.

Cont..

- ❖ In general, a **functional dependency** is a relationship among attributes. In relational databases, we can have a determinant that governs one other attribute or several other attributes.
- ❖ Functional dependency is represented by an arrow sign $\rightarrow$ that is, $X \rightarrow Y$, where X functionally determines Y . The left-hand side attributes determine the values of attributes on the right-hand side.
- ❖ $X \rightarrow Y$ holds if two tuples have the same value for X , they must have the same value for Y .
- ❖ $A \rightarrow B$ is read as; B is functionally dependent on A

Cont..

❖ We also often express this idea by saying that:

- ✓ "A determines B," or that
- ✓ "B is a function of A," or that
- ✓ "A functionally governs B."

Example

❖ **Dinner**

Type of wine

❖ *Meat*

Red

❖ *Fish*

White

❖ *Cheese*

Rose

❖ Since the type of Wine served depends on the type of Dinner, we say Wine is functionally dependent on Dinner.

❖ *Dinner* → *Wine*

Types of Functional Dependency

Partial Dependency:

- ❖ If an attribute which is not a member of **the primary key** is dependent on some **part of the primary key** (if we have composite primary key) then that attribute is partially functionally dependent on the primary key.
- ❖ Let $\{A, B\}$ is the Primary Key and C is no key attribute.
 - ❖ Then if $\{A, B\} \rightarrow C$ and $B \rightarrow C$ or $A \rightarrow C$
 - ❖ Then C is partially functionally dependent on $\{A, B\}$

Full Dependency:

- ❖ If an attribute which is **not a member of the primary key** is not dependent on **some part of the primary key** but the **whole key** (if we have composite primary key) then that attribute is fully functionally dependent on the primary key.
- ❖ Let $\{A, B\}$ is the Primary Key and C is no key attribute
 - ❖ Then if $\{A, B\} \rightarrow C$ and $B \rightarrow C$ and $A \rightarrow C$ does not hold (if B cannot determine C and A cannot determine C).
 - ❖ Then C Fully functionally dependent on $\{A, B\}$.

Transitive Dependency:

- ❖ In mathematics and logic, a transitive relationship is a relationship of the following form: "If A implies B, and if also B implies C, then A implies C."
- ❖ Example:
 - ❖ *If Mr. X is a Human, and if every Human is an Animal, then Mr. X must be an Animal.*

Cont..

❖ Generalized way of describing transitive dependency is that:

❖ If A functionally governs B, AND

❖ If B functionally governs C

❖ THEN A functionally governs C

❖ Provided that neither C nor B determines A i.e. ($B \not\rightarrow A$ and $C \not\rightarrow A$)

❖ In the normal notation:

❖ $\{(A \rightarrow B) \text{ AND } (B \rightarrow C)\} \implies A \rightarrow C$ provided that $B \not\rightarrow A$ and $C \not\rightarrow A$

Normalization

- ❖ **Database normalization** is a series of steps followed to obtain a database design that allows for consistent storage and efficient access of data in a relational database. These steps reduce data redundancy and the risk of data becoming inconsistent.
- ❖ It is the process of identifying the logical associations between data items and designing a database that will represent such associations but without suffering the update anomalies which are;
 - ❖ **Insertion Anomaly** – adding new rows forces user to create duplicate data
 - ❖ **Deletion Anomaly** – deleting a row may cause loss of other data representing completely different facts
 - ❖ **Modification Anomaly** – changing data in a row forces changes to other rows because of duplication

Steps of Normalization

- ❖ We have various levels or steps in normalization called Normal Forms.
- ❖ A table in a relational database is said to be in a certain normal form if it satisfies certain constraints.

❖ **Normalization towards a logical design consists of the following steps:**

❖ ***Un-Normalized Form:***

❖ Identify all data elements

❖ ***First Normal Form:***

❖ Find the key with which you can find all data

❖ ***Second Normal Form:***

❖ Remove partial dependencies.

❖ Make all data dependent on the whole key.

❖ ***Third Normal Form***

❖ Remove non-key dependencies.

❖ Make all data dependent on nothing but the key.

❖ For most practical purposes, databases are considered normalized if they adhere to third normal form.

First Normal Form (1NF)

- ❖ Requires that all column values in a table are atomic (e.g., a number is an atomic value, while a list or a set is not).
- ❖ We have two ways of achieving this:
 - ❖ Putting each repeating group into a separate table and connecting them with a primary key-foreign key relationship
 - ❖ Moving this repeating group to a new row by repeating the common attributes.
 - ❖ If so then Find the key with which you can find all data.

Definition: a table (relation) is in 1NF

If

- ❖ There are no duplicated rows in the table. Unique identifier
- ❖ Each cell is single-valued (i.e., there are no repeating groups).
- ❖ Entries in a column (attribute, field) are of the same kind.

❖ Example for First Normal form (1NF)

Not normalized

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn	Dlocations
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

IN FIRST NORMAL FORM (1NF)

- ❖ Remove all repeating groups.
- ❖ Distribute the multi-valued attributes into different rows and identify a unique identifier for the relation so that it can be said is a relation in relational database.

DEPARTMENT

Dname	<u>Dnumber</u>	Dmgr_ssn	<u>Dlocation</u>
Research	5	333445555	Bellaire
Research	5	333445555	Sugarland
Research	5	333445555	Houston
Administration	4	987654321	Stafford
Headquarters	1	888665555	Houston

Second Normal form 2NF

- ❖ No partial dependency of a non-key attribute on part of the primary key.
- ❖ A relation schema R is in 2NF if every nonprime attribute A in R is fully functionally dependent on the primary key of R.
- ❖ This will result in a set of relations with a level of Second Normal Form.
- ❖ Any table that is in 1NF and has a single-attribute (i.e., a non-composite) primary key is automatically in 2NF.
- ❖ **Definition: a table (relation) is in 2NF**

If

- ❖ It is in 1NF and
- ❖ If all non-key attributes are dependent on the entire primary key.
i.e. no partial dependency.

Example for 2NF:

EMP_PROJ

<u>EmpID</u>	<u>EmpName</u>	<u>ProjNo</u>	<u>ProjName</u>	<u>ProjLoc</u>	<u>ProjFund</u>	<u>ProjMangID</u>	Incentive
--------------	----------------	---------------	-----------------	----------------	-----------------	-------------------	-----------

EMP_PROJ rearranged

<u>EmpID</u>	<u>ProjNo</u>	<u>EmpName</u>	<u>ProjName</u>	<u>ProjLoc</u>	<u>ProjFund</u>	<u>ProjMangID</u>	Incentive
--------------	---------------	----------------	-----------------	----------------	-----------------	-------------------	-----------

- ❖ This schema is in its 1NF since we don't have any repeating groups or attributes with multi-valued property.
- ❖ To convert it to a 2NF we need to remove all partial dependencies of non key attributes on part of the primary key.

Cont..

- ❖ $\{EmpID, ProjNo\} \rightarrow EmpName, ProjName, ProjLoc, ProjFund, ProjMangID, Incentive$
- ❖ But in addition to this we have the following dependencies
 - ✓ FD1: $\{EmpID\} \rightarrow EmpName$
 - ✓ FD2: $\{ProjNo\} \rightarrow ProjName, ProjLoc, ProjFund, ProjMangID$
 - ✓ FD3: $\{EmpID, ProjNo\} \rightarrow Incentive$

EMPLOYEE

<u>EmpID</u>	EmpName
--------------	---------

PROJECT

<u>ProjNo</u>	ProjName	ProjLoc	ProjFund	ProjMangID
---------------	----------	---------	----------	------------

EMP_PROJ

<u>EmpID</u>	<u>ProjNo</u>	Incentive
--------------	---------------	-----------

Third Normal Form (3NF)

- ❖ Eliminate columns dependent on another non-Primary Key.
- ❖ If attributes do not contribute to a description of the key, remove them to a separate table.
- ❖ This level avoids update and delete anomalies.
- ❖ ***Definition: a Table (Relation) is in 3NF***

If

- ❖ It is in 2NF and
- ❖ There are no transitive dependencies between a primary key and non-primary key attributes.

Example for (3NF)

❖ Assumption: Students of same batch live in one building or dormitory

STUDENT

<u>StudID</u>	<u>Stud F Name</u>	<u>Stud L Name</u>	<u>Dept</u>	<u>Year</u>	<u>Dormitory</u>
125/97	Abebe	Mekuria	Info Sc	1	401
654/95	Lemma	Alemu	Geog	3	403
842/95	Chane	Kebede	CompSc	3	403
165/97	Alem	Kebede	InfoSc	1	401
985/95	Almaz	Belay	Geog	3	403

Cont..

- ❖ This schema is in its 2NF since the primary key is a single attribute.
- ❖ Let's take StudID, Year and Dormitory and see the dependencies.
 - ❖ $\text{StudID} \rightarrow \text{Year}$ AND $\text{Year} \rightarrow \text{Dormitory}$
 - ❖ And Year cannot determine StudID and Dormitory can not determine StudID
 - ❖ Then transitively $\text{StudID} \rightarrow \text{Dormitory}$
- ❖ To convert it to a 3NF we need to remove all transitive dependencies of non key attributes on another non-key attribute.
- ❖ The non-primary key attributes, dependent on each other will be moved to another table and linked with the main table using Candidate Key-Foreign Key relationship.

Cont...

STUDENT

<u>StudID</u>	<u>Stud F_Name</u>	<u>Stud L_Name</u>	<u>Dept</u>	<u>Year</u>
125/97	Abebe	Mekuria	Info Sc	1
654/95	Lemma	Alemu	Geog	3
842/95	Chane	Kebede	CompSc	3
165/97	Alem	Kebede	InfoSc	1
985/95	Almaz	Belay	Geog	3

DORM

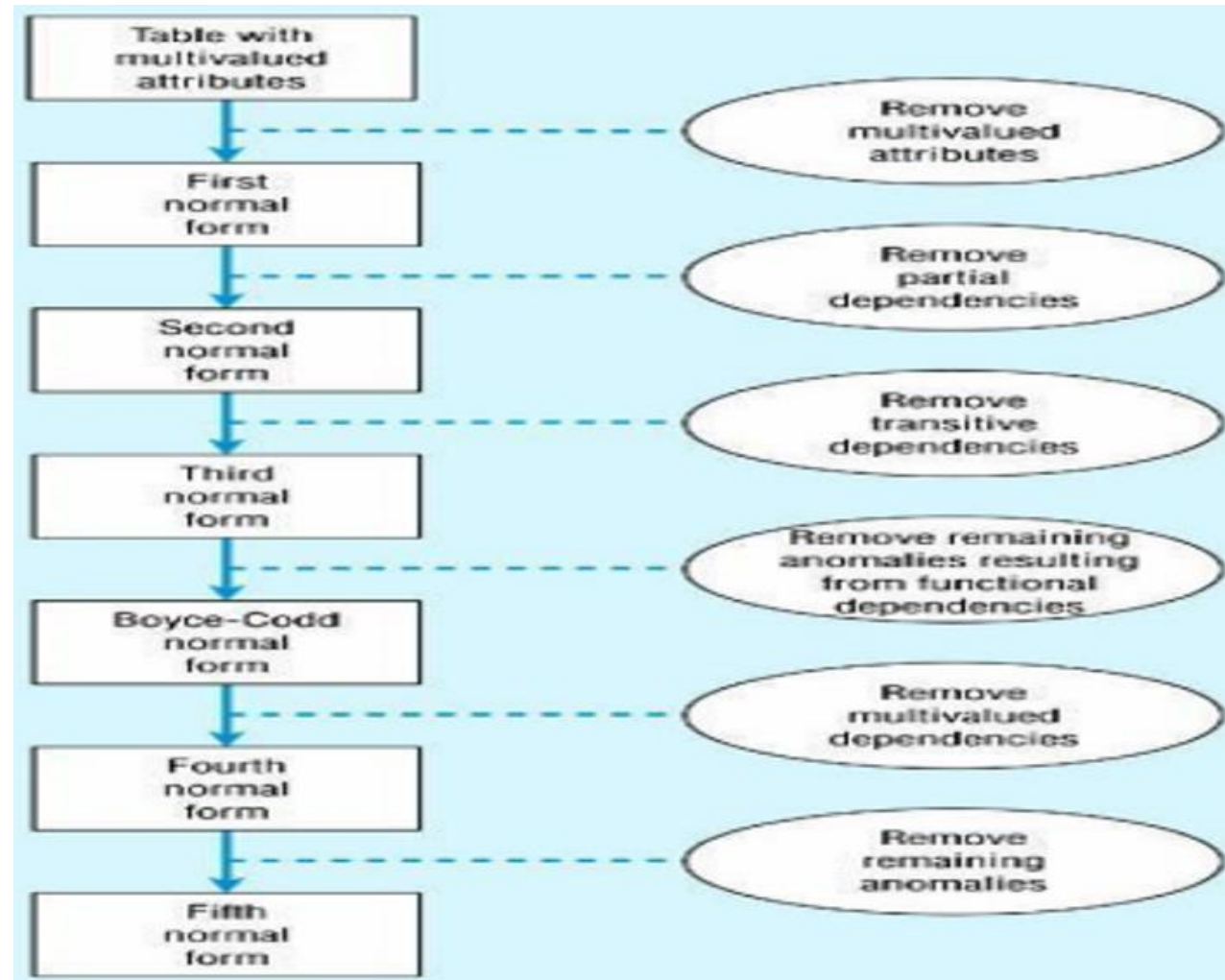
<u>Year</u>	<u>Dormitory</u>
1	401
3	403

Cont..

- ❖ Generally, even though there are other four additional levels of normalization, a table is said to be normalized if it reaches 3NF.
- ❖ A database with all tables in the 3NF is said to be normalized database.
- ❖ NB: to remember the rationale for normalization up to 3NF:
 - ✓ No repeating or redundancy fields in the table.
 - ✓ The fields depend upon the key: the table should solely depend on the key.
 - ✓ The whole key: no partial key dependency.
 - ✓ And nothing but the key: no inter data dependency.

Cont..

❖ Graphical illustration of different phases of normalization



Additional examples for normal forms

Sales Records:

Cust Name	Item	Shipping Address	Newsletter	Supplier	Supplier Phone	Price
Alan Smith	Xbox One	35 Palm St, Miami	Xbox News	Microsoft	(800) BUY-XBOX	250
Roger Banks	PlayStation 4	47 Campus Rd, Boston	PlayStation News	Sony	(800) BUY-SONY	300
Evan Wilson	Xbox One, PS Vita	28 Rock Av, Denver	Xbox News, PlayStation News	Wholesale	Toll Free	450
Alan Smith	PlayStation 4	47 Campus Rd, Boston	PlayStation News	Sony	(800) BUY-SONY	300

- 1st Normal Form**
- Each cell to be Single valued
 - Entries in a column are same type
 - Rows uniquely identified - Add Unique ID, or Add more columns to make unique
(Note: The order of the rows and the order of the columns are irrelevant)

Primary Key

Cust ID	Cust Name	Item	Shipping Address	Newsletter	Supplier	Supplier Phone	Price
at_smith	Alan Smith	Xbox One	35 Palm St, Miami	Xbox News	Microsoft	(800) BUY-XBOX	250
roger25	Roger Banks	PlayStation 4	47 Campus Rd, Boston	PlayStation News	Sony	(800) BUY-SONY	300
wilson44	Evan Wilson	Xbox One	28 Rock Av, Denver	Xbox News	Microsoft	(800) BUY-XBOX	250
wilson44	Evan Wilson	PS Vita	28 Rock Av, Denver	PlayStation News	Sony	(800) BUY-SONY	200
am_smith	Alan Smith	PlayStation 4	47 Campus Rd, Boston	PlayStation News	Sony	(800) BUY-SONY	300

Cont..

2nd Normal Form - All attributes (Non-Key Columns) dependent on the key

Primary Key

Cust ID	Cust Name	Shipping Address	Newsletter
at_smith	Alan Smith	35 Palm St, Miami	Xbox News
roger25	Roger Banks	47 Campus Rd, Boston	PlayStation News
wilson44	Evan Wilson	28 Rock Av, Denver	Xbox News
wilson44	Evan Wilson	28 Rock Av, Denver	PlayStation News
am_smith	Alan Smith	47 Campus Rd, Boston	PlayStation News

Primary Key

Item	Supplier	Supplier Phone	Price
Xbox One	Microsoft	(800) BUY-XBOX	250
PlayStation 4	Sony	(800) BUY-SONY	300
PS Vita	Sony	(800) BUY-SONY	200

Primary Key

Primary Key

Cust ID	Item
at_smith	Xbox One
roger25	PlayStation 4
wilson44	Xbox One
wilson44	PS Vita
am_smith	PlayStation 4

Cont..

3rd Normal Form - All Fields (columns) can be determined Only by the Key in the table and no other column

Primary Key

Cust ID	Cust Name	Shipping Address	Newsletter
at_smith	Alan Smith	35 Palm St, Miami	Xbox News
roger25	Roger Banks	47 Campus Rd, Boston	PlayStation News
wilson44	Evan Wilson	28 Rock Av, Denver	Xbox News
wilson44	Evan Wilson	28 Rock Av, Denver	PlayStation News
am_smith	Alan Smith	47 Campus Rd, Boston	PlayStation News

Primary Key

Foreign Key

Item	Supplier	Price
Xbox One	Microsoft	250
PlayStation 4	Sony	300
PS Vita	Sony	200

Primary Key

Primary Key

Cust ID	Item
at_smith	Xbox One
roger25	PlayStation 4
wilson44	Xbox One
wilson44	PS Vita
am_smith	PlayStation 4

Primary Key

Supplier	Supplier Phone
Microsoft	(800) BUY-XBOX
Sony	(800) BUY-SONY

End of Chapter 4

Thank You!